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AK-MCS: An active learning reliability method combining Kriging and Monte Carlo Simulation

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ABSTRACT

An important challenge in structural reliability is to keep to a minimum the number of calls to the numerical models. Engineering problems involve more and more complex computer codes and the evaluation of the probability of failure may require very time-consuming computations. Metamodels are used to reduce these computation times. To assess reliability, the most popular approach remains the numerous variants of response surfaces. Polynomial Chaos [1] and Support Vector Machine [2] are also possibilities and have gained considerations among researchers in the last decades. However, recently, Kriging, originated from geostatistics, have emerged in reliability analysis. Widespread in optimisation, Kriging has just started to appear in uncertainty propagation [3] and reliability [4,5] studies. It presents interesting characteristics such as exact interpolation and a local index of uncertainty on the prediction which can be used in active learning methods. The aim of this paper is to propose an iterative approach based on Monte Carlo Simulation and Kriging metamodel to assess the reliability of structures in a more efficient way. The method is called AK-MCS for Active learning reliability method combining Kriging and Monte Carlo Simulation. It is shown to be very efficient as the probability of failure obtained with AK-MCS is very accurate and this, for only a small number of calls to the performance function. Several examples from literature are performed to illustrate the methodology and to prove its efficiency particularly for problems dealing with high non-linearity, non-differentiability, non-convex and non-connex domains of failure and high dimensionality.

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1. Introduction

Reliability aims at determining the probability of failure of a system by considering its inputs as random. The performance function $G(\mathbf{x})$ characterizes the response of the system: a negative value means that the failure occurs, a positive implies that the system is safe in this configuration. The border between the negative and positive domains is called the limit state ($G(\mathbf{x}) = 0$). To assess the probability of failure, several methods exist. The most common of them remains the Monte Carlo Simulation. It consists of a random population of size n_{MC} for which the performance function is evaluated. An estimation of the probability of failure P_f is then obtained as the ratio between $n_{G\leqslant 0}$, the number of samples giving a negative or null G values and the size of the sampling n_{MC}:

$$P_f = \frac{n_{G\leqslant 0}}{n_{MC}} \tag{1}$$

The number of evaluations n_{MC} can be extremely important as it depends on the expected probability of failure. Indeed, a very weak probability of failure requires a large random population to ascertain the estimation. To quantify the uncertainty on the probability of failure, its coefficient of variation is calculated as:

$$\mathrm{C.O.V}_{P_f} = \sqrt{\frac{1 - P_f}{P_f\, n_{MC}}} \tag{2}$$

Crude Monte Carlo Simulations are easy to implement and are used in a large domain of applications. However, the assessment of the probability of failure requires the evaluation of the performance function for the whole population. This can easily be done on analytical systems as the computational demand can be neglected but it can become impossible with complex computer codes such as Finite Element Models for example. Engineering models have become extremely time-consuming process despite the improvements in computer technology. Therefore, it became necessary to develop methods to approximate the probability of failure with fewer calls to the expensive performance function. FORM and SORM are two elementary approaches [6,7]. They are based on finding the named "design point" and present great advantages

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for very weak probabilities of failure, yet they cannot be applied to all problems (especially with high non-linear or complex limit states in the standard Gaussian space). This led the path to the use of metamodels. These models aim at approximating the performance function with a strategic design of experiments (DoE) to obtain, for a less expensive computational demand, a sufficiently accurate prediction of the performance function's sign. Response surfaces are the most popular methods. They are used for their speed but they are limited in global interpolation accuracy. To avoid this problem, Polynomial Chaos can be used [1,8]. It corresponds to a response surface in a particular base (Hermite or other polynomials). However, the definitions of the design of experiments and of the polynomial's degrees are tricky [9]. Furthermore, the accuracy evaluation requires cross validation. More recently, Support Vector Machine methods have also gained considerations with their great effectiveness for reliability studies [2,10].

Aside from these metamodels, a stochastic approach has been intensively investigated: Kriging. Developed for geostatistics in the fifties and sixties by Krige and then by Matheron [11], this method gained considerations in the computer experiments field in the eighties. It presents several but interesting differences with the other metamodels. First, Kriging is an exact interpolation method, *i.e.* the prediction in a point belonging to the design of experiments is the exact value of the performance function in this point. Furthermore, thanks to its stochastic propriety, Kriging provides not only predicted values in any points, but also estimations of the local variance of the predictions. This variance defines the local uncertainty on the prediction and is called the Kriging variance. It is understood in the way that the higher the variance, the less certain the prediction. Thanks to this variance, Kriging has been intensively used in optimisation problems in the nineties with active learning methods such as Efficient Global Optimisation (EGO) [12]. Active learning means that the Kriging model is updated by adding a new point to the design of experiments, this point being selected for its expected improvement on the Kriging model. In this domain, EGO was a major step forward.

The applications of Kriging to structural reliability problems are rather recent. The precursor work seems to have been proposed by Romero et al. [3] who compare Kriging with polynomial regression and finite-element interpolation on progressive lattice-samplings with analytical functions. The results show that none of the methods is more efficient than the others. Following this, Kaymaz [4] proposes a method to perform structural reliability analysis and compares it to classic response surface methods. His method is based on the MATLAB toolbox DACE [13,14] and consists of finding the design point. His conclusions are that Kriging metamodel does not greatly improve the reliability results compared to quadratic response surface methods' results, unless Kriging parameters are well chosen. Following this, structural reliability problems are also investigated using Kriging in [15]. These methods are based on progressive design of experiments but they cannot be defined as active learning methodologies. Indeed, the design of experiments is not improved by learning from all the data supplied by Kriging, *i.e.* Kriging prediction and variance. This was noticed by Bichon et al. [5] who propose a fully active learning method to perform reliability. Inspired by EGO and the Kriging contour estimation method of Ranjan et al. [16], this method is called Efficient Global Reliability Analysis (EGRA). The active learning approach is defined thanks to a learning function called the Expected Feasibility Function (EFF) which provides an indication of how well the true value of the performance function in a point can be expected to satisfy the constraint $G(\mathbf{x}) = 0$. EFF is expressed as a function of the Kriging data (local prediction and local variance). More information about it is given Section 3.3. This method consists of a sequential algorithm starting with $\frac{(n+1)(n+2)}{2}$ samples (n being the number of random variables) in the initial design of experiments defined by using Latin Hypercube sampling over the bounds ± five standard deviations. From this design, the Kriging model is constructed and the point in space which maximises the learning function EFF (learning criterion) is searched using the DIRECT global optimisation algorithm [17] (this algorithm is preferred to the BRANCH-AND-BOUND [18] one used in the Kriging contour estimation method [16] which they see as too expensive). This point is then evaluated on the performance function and added to the design of experiments. This is done until some stopping condition is satisfied. The surrogate model is then considered accurate enough to calculate the probability of failure using Importance Sampling (IS). The results show that the method is really efficient. However, it can be noticed that in high dimensions problems, the size of the initial design of experiments is extremely important. Furthermore, the method approximates the limit state in the whole design space and therefore, even in regions, where configurations show very weak densities of probability and have negligible effects on the probability of failure.

To avoid that, this paper proposes to develop an Active learning method which combines Kriging and Monte Carlo Simulation to separate predicted negative and positive $\widehat{G}$ values of a Monte Carlo population. The idea is to perform a Monte Carlo Simulation without evaluating the whole population on G. The sign in each point is obtained thanks to the predictions thanks to a Kriging model based on a few evaluated points. The first stage of the method consists of generating a Monte Carlo population. An initial design of experiments of very small size is selected among this population and a learning function is computed for all the points. The best next point to include in the design of experiments is selected by this function. The probability of failure is estimated at each step thanks to the Kriging predictions of the Monte Carlo population. This method enables to evaluate points near the limit state to improve the accuracy of the metamodel and above all, it enables to focus only on the points having sufficiently high densities of probability to have a significant impact on the probability of failure. The approximation of the limit state is then very accurate in the Monte Carlo population. This method gives at the same time a Kriging estimation of the probability of failure and its coefficient of variation without requiring the expensive evaluation of the whole Monte Carlo population. It is named AK-MCS as it is an Active learning method combining Kriging and Monte Carlo Simulation. It must be seen as a modification of the crude Monte Carlo Simulation.

This article is framed in four sections. Kriging theory is presented Section 2 to show the real interest in using it in computer codes. Following this, the proposed method (AK-MCS) is detailed Section 3 as a solution to the current existing methodologies' problems. Its concept is defined as different from previous works. Section 4 proposes several examples to validate AK-MCS. It is compared to Monte Carlo Simulation and results of the literature. AK-MCS is shown to be very efficient as the number of calls to the performance function is relatively weak compared to the other approaches.

2. Kriging theory: a reminder

Kriging is based on the idea that the performance function $G(\mathbf{x})$ is seen as the realisation of a stochastic field $\mathcal{G}(\mathbf{x})$ [11]. The first step of Kriging consists of defining this stochastic field with its parameters according to a design of experiments. Then, the Best Linear Unbiased Predictor (BLUP) is used to estimate the value in a given point. The model [19] for $\mathcal{G}(\mathbf{x})$ is given as:

$$\mathcal{G}(\mathbf{x}) = F(\mathbf{x}, \boldsymbol{\beta}) + z(\mathbf{x}) \tag{3}$$

where:

- $F(\mathbf{x},\boldsymbol{\beta})$ is the deterministic part which gives an approximation of the response in mean. It represents the trend of Kriging and corresponds to a regression model that can be written:

$$F(\mathbf{x},\boldsymbol{\beta}) = \mathbf{f}(\mathbf{x})^t\boldsymbol{\beta} \tag{4}$$

  with $\mathbf{f}(\mathbf{x})^t = \{f_1(\mathbf{x}),\ldots,f_k(\mathbf{x})\}$ the basis functions and $\boldsymbol{\beta}^t = \{\beta_1,\ldots,\beta_k\}$ the vector of regression coefficients. In this paper, ordinary Kriging is selected which means that $F(\mathbf{x},\boldsymbol{\beta})$ is a scalar to determine: $F(\mathbf{x},\boldsymbol{\beta}) = F(\mathbf{x},\beta) = \beta$. All the following equations are now based on ordinary Kriging theory [11].
- $z(\mathbf{x})$ is a stationary Gaussian process with zero mean and covariance between two points of space $\mathbf{x}$ and $\mathbf{w}$ defined by:

$$\mathrm{cov}(z(\mathbf{x}),z(\mathbf{w})) = \sigma_z^2 R_\theta(\mathbf{x},\mathbf{w}) \tag{5}$$

  where σ_z^2 is the process variance and R_θ the correlation function defined by its set of parameters $\boldsymbol{\theta}$.

Several models exist to define the correlation function. However, in this paper, the anisotropic squared-exponential function (also called anisotropic Gaussian model) is selected. The anisotropy aspect is defined by the fact that $\boldsymbol{\theta}$ is a vector of length n which corresponds to the number of random variables. The correlation model can be formulated by:

$$R_\theta(\mathbf{x},\mathbf{w}) = \prod_{i=1}^{n} \exp[-\theta_i(x_i - w_i)^2] \tag{6}$$

where x_i and w_i are the ith coordinates of the points $\mathbf{x}$ and $\mathbf{w}$ and θ_i is a scalar which gives the multiplicative inverse of the correlation length in the ith direction. An anisotropic correlation function is preferred here as in reliability studies, the random variables are often of different natures.

Given a design of experiments $[\mathbf{x}^{(1)},\ldots,\mathbf{x}^{(p)}]$, with $\mathbf{x}^{(i)} \in \mathbb{R}^n$ the ith experiment, and $\mathbf{Y}$ with $Y^{(i)} = G(\mathbf{x}^{(i)}) \in \mathbb{R}$, the scalars β and σ_z^2 are estimated according to [12] by:

$$\hat{\beta} = \left(\mathbf{1}^t\mathbb{R}_\theta^{-1}\mathbf{1}\right)^{-1}\mathbf{1}^t\mathbb{R}_\theta^{-1}\mathbf{Y} \tag{7}$$

$$\hat{\sigma_z}^2 = \frac{1}{p}(\mathbf{Y}-\beta\mathbf{1})^t\mathbb{R}_\theta^{-1}(\mathbf{Y}-\beta\mathbf{1}) \tag{8}$$

where $R_{\theta_{ij}} = R_\theta(\mathbf{x}^{(i)},\mathbf{x}^{(j)})$ is the matrix of correlation between each pair of points of the design of experiments and $\mathbf{1}$ the vector filled with 1 of length p.

However, as $\hat{\beta}$ and $\hat{\sigma_z}^2$ in Eqs. (7) and (8) depend on the correlation parameters θ_i through the matrix $\mathbb{R}_\theta$, it is first required to obtain them using maximum likelihood estimation:

$$\boldsymbol{\theta} = \arg\min_{\boldsymbol{\theta}} (\det \mathbb{R}_\theta)^{\frac{1}{p}}\,\hat{\sigma_z}^2 \tag{9}$$

The next step consists of predicting the response of $G(\mathbf{x})$ in a given unknown point. At such a point as $\mathbf{x}$, the Best Linear Unbiased Predictor (BLUP) $\widehat{G}(\mathbf{x})$ of $\mathcal{G}(\mathbf{x})$ is computed [11]:

$$\widehat{G}(\mathbf{x}) = \beta + \mathbf{r}(\mathbf{x})\mathbb{R}_\theta^{-1}(\mathbf{Y}-\beta\mathbf{1}) \tag{10}$$

where $\mathbf{r}(\mathbf{x}) = \{R_\theta(\mathbf{x},\mathbf{x}^{(i)})\}_{i=1,\ldots,p}$.

The Kriging variance $\sigma^2_{\widehat{G}}(\mathbf{x})$ is defined as the minimum of the mean squared error between $\widehat{G}(\mathbf{x})$ and $G(\mathbf{x})$. It can be expressed as the following analytical function:

$$\sigma^2_{\widehat{G}}(\mathbf{x}) = \sigma_z^2\left(1 + u(\mathbf{x})^t\left(\mathbf{1}^t\mathbb{R}_\theta^{-1}\mathbf{1}\right)^{-1}u(\mathbf{x}) - \mathbf{r}(\mathbf{x})^t\mathbb{R}_\theta^{-1}\mathbf{r}(\mathbf{x})\right) \tag{11}$$

where $u(\mathbf{x}) = \mathbf{1}^t\mathbb{R}_\theta^{-1}\mathbf{r}(\mathbf{x}) - 1$.

Kriging is an exact interpolation method. The prediction $\widehat{G}(\mathbf{x}^{(i)})$ in a point $\mathbf{x}^{(i)}$ of the design of experiments is exact, *i.e.* $\widehat{G}(\mathbf{x}^{(i)}) = G(\mathbf{x}^{(i)})$. Therefore, the Kriging variance is null in these points and it becomes important in unexplored areas. This enables to quantify the uncertainty of local predictions with an easily computable analytical function. These properties are interesting in reliability studies and metamodels as the Kriging variance represents a good index to improve a design of experiments.

3. Proposed method: AK-MCS

3.1. Principles of the methodology

The method that we propose here is named AK-MCS. It consists of an Active learning reliability method combining Kriging and Monte Carlo Simulation. The active learning principles for reliability with Kriging are proposed in the previous works [5,16]. However, the concept here is different. In fact, instead of approximating the limit state in the whole space and therefore making expensive evaluations of G for points with very weak densities of probability, it is preferred here to focus on a Monte Carlo population. AK-MCS classifies a Monte Carlo population of n_{MC} points without evaluating n_{MC} times the true performance function. To avoid these evaluations, Kriging is used for its exact interpolation characteristic and for its interesting proprieties in active learning methods. Indeed, as shown in introduction for EFF, its stochastic nature enables to define a learning function to select among the n_{MC} points the best next point to evaluate on the performance function. In this section, AK-MCS, which must be seen as a modification of the classic Monte Carlo Simulation, is introduced. Then, two learning functions are presented : EFF coming from the EGRA method [5] and a more suitable learning function for AK-MCS that we propose to name U.

3.2. AK-MCS

The method proposed is given Fig. 1. It consists of 10 stages:

1. *Generation of a Monte Carlo population in the design space.* It is named S and is composed of n_{MC} points. At this stage, none of them is evaluated on the performance function. They represent candidate points to be evaluated if the active learning requires it. This population remains the same during the whole process of learning in AK-MCS unless Stage 9 is reached.
2. *Definition of the initial design of experiments (DoE).* To perform Kriging, a design of experiments is required. Here, it consists of a random selection of N_1 points among the population S. These points are evaluated on the performance function and are used as the initial design of experiments for the Kriging model. According to our experience, a dozen of points are enough. The initial design of experiments is preferred to be defined small and to add step by step only the point that is improving the most the metamodel. In the previous cited papers [5,16], the initial design of experiments is much larger as they grow with the dimension of the problem.
3. *Computation of the Kriging model according to the design of experiments.* This stage is performed with the toolbox DACE, as this tool has been used in several Kriging reference articles [4,20–22]. The correlation model is chosen Gaussian and the regression model is taken constant (Ordinary Kriging), as mentioned Section 2.
4. *Prediction by Kriging and estimation of the probability of failure.* First, Kriging predictions $\widehat{G}(\mathbf{x}^{(i)})$ for $i = 1,\ldots,n_{MC}$ with Eq. (10) are obtained thanks to DACE. Then, the probability of failure is estimated with the signs of these predictions. It is obtained by the ratio of the points in the population S with a negative or null Kriging prediction and the total number of points in S following Eq. (1):

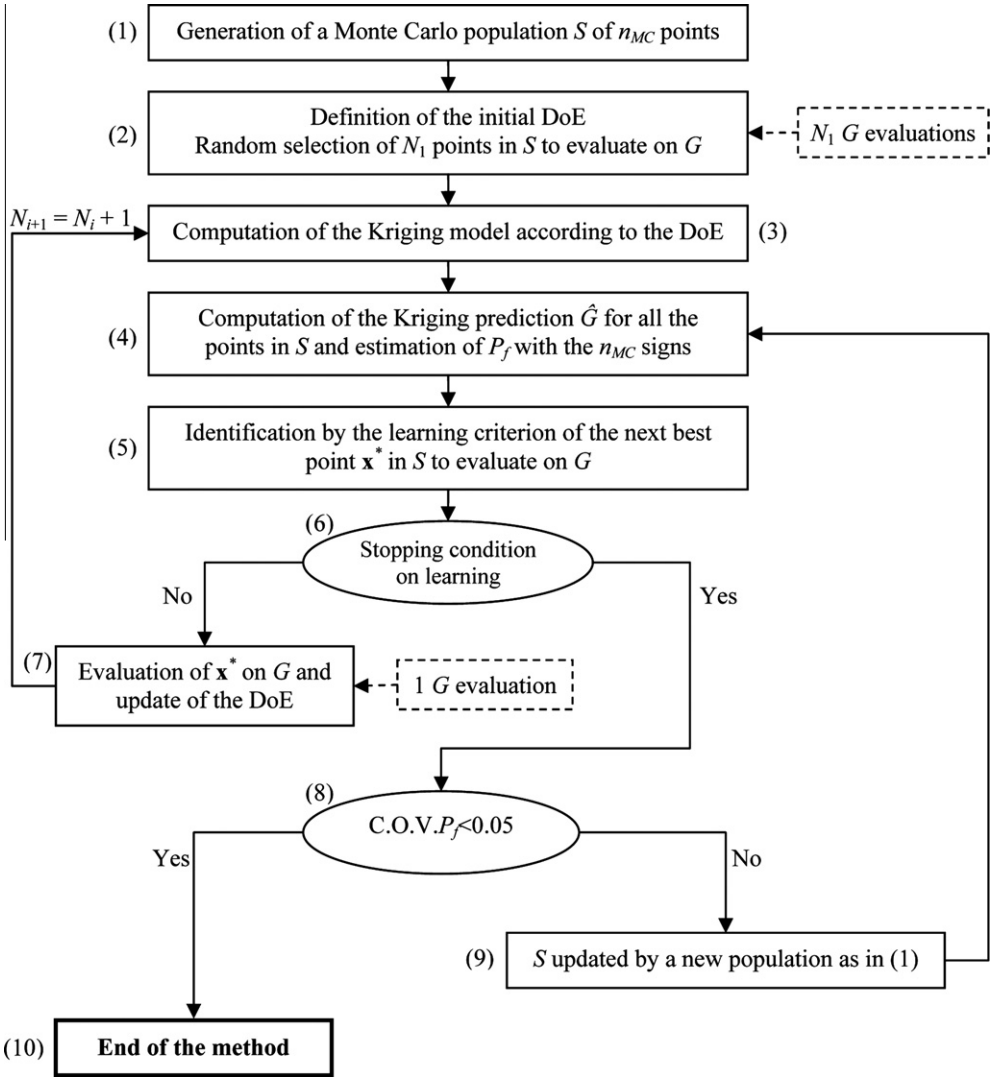


Fig. 1. AK-MCS flowchart.

$$\widehat{P_f} = \frac{n_{\widehat{G\leqslant 0}}}{n_{MC}} \tag{12}$$

5. *Identification of the best next point in S to evaluate on the performance function.* This stage requires the implementation of what we name the learning function. This function is computed for each point of S. As it depends on the Kriging prediction and variance, the Kriging variances $\sigma^2_{\widehat{G}}(\mathbf{x}^{(i)})$ for $i = 1,\ldots,n_{MC}$ are first computed with Eq. (11). In this article, we use two learning functions. EFF is proposed in [5] and we implement a new function named U. The best next point $\mathbf{x}^*$ in S to evaluate on the performance function is then identified according to the learning function values observed for all the points of S. The criterion to identify $\mathbf{x}^*$ is named the learning criterion as it depends on the choice of the learning function. It is discussed Section 3.3 for each learning function.
6. *Stopping condition on learning.* Once the best next point $\mathbf{x}^*$ in S to evaluate has been identified with the learning criterion, its corresponding learning function's value is compared to a stopping one. This is referred as the stopping condition. Once again, it depends on the choice of the learning function discussed Section 3.3.
7. *Update of the previous design of experiments with the best point.* If the stopping condition Stage 6 is not satisfied, the learning carries on and the best point $\mathbf{x}^*$ in S is evaluated on the true performance function. Following this, it is added to the design of experiments: $N_{i+1} = N_i + 1$. The method, then, goes back to Stage 3 to compute the new Kriging model with the updated design of experiments composed of N_{i+1} points.
8. *Computation of the coefficient of variation of the probability of failure.* If the stopping condition Stage 6 is satisfied, the learning is stopped and the metamodel is said to be accurate enough on the performance function's signs of the n_{MC} points. The next stage consists of checking if the Monte Carlo population S is sufficiently large to obtain a low coefficient of variation on the Kriging estimation of the probability of failure (Stage 4). Here, a coefficient of variation below 5% is seen as acceptable. It is calculated the same way than Eq. (2) but on the Kriging results:

$$\text{C.O.V}_{\widehat{P_f}} = \sqrt{\frac{1-\widehat{P_f}}{\widehat{P_f}\, n_{MC}}} \tag{13}$$

9. *Update of the population.* If the coefficient of variation is found to be too high, S is increased with new points coming from another Monte Carlo population (generated like Stage 1). AK-MCS goes back to Stage 4 to predict the new population and the active learning method carries on until the stopping condition is met again. It is important to note that no information about previous performance function's evaluations is lost.
10. *End of AK-MCS.* If the coefficient of variation of $\widehat{P_f}$ is low enough, AK-MCS stops and the last estimation of the probability of failure is considered as the result of the method.

The advantage of AK-MCS is that it enables to focus only on the configurations which are likely to happen for a certain level of failure. It separates the negative and the positive predicted values like a Monte Carlo Simulation would do on the performance function. Furthermore, the points that are added to the design of experiments are only among those already included in the initial Monte Carlo population and the probability of failure is computed at each iteration. It becomes easier to follow the convergence of the probability of failure in terms of the number of calls to the performance function. Moreover, the DIRECT [5] or BRANCH-AND-BOUND [16] algorithms do not need to be implemented as the learning is not performed on the whole design space but only among the population. Even if the number of predictions by Kriging in the present method can be important as the whole Monte Carlo population is estimated, the computation time of the predictions can be neglected compared to the time required to evaluate the performance function (particularly if the performance function is calculated by a complex computer code). Furthermore, the DIRECT and BRANCH-AND-BOUND algorithms get expensive to perform in high dimensions. Here, high dimension is not a problem as the Monte Carlo population remains the same throughout iterations and only depends on the probability of failure to reach.

3.3. Learning functions

The learning functions are essential to AK-MCS. They are used to determine the best next point to evaluate on the performance function. They represent the active learning aspect of AK-MCS. In introduction, EFF is mentioned. More information about it is given in the first paragraph of this section. Following this, a more suitable function named U for AK-MCS is proposed.

3.3.1. Expected feasibility function (EFF)

EFF comes from the EGRA method [5]. Its general formulation provides an indication of how well the true value of the performance function in a point $\mathbf{x}$ is expected to satisfy the equality constraint $G(\mathbf{x}) = a$ over a region defined by $a \pm \epsilon$. It reads:

$$\begin{aligned}\mathrm{EFF}(\mathbf{x}) = (&\widehat{G}(\mathbf{x}) \\ &- a)\left[2\Phi\left(\frac{a-\widehat{G}(\mathbf{x})}{\sigma_{\widehat{G}}(\mathbf{x})}\right) - \Phi\left(\frac{(a-\epsilon)-\widehat{G}(\mathbf{x})}{\sigma_{\widehat{G}}(\mathbf{x})}\right) - \Phi\left(\frac{(a+\epsilon)-\widehat{G}(\mathbf{x})}{\sigma_{\widehat{G}}(\mathbf{x})}\right)\right] \\ &- \sigma_{\widehat{G}}(\mathbf{x})\left[2\phi\left(\frac{a-\widehat{G}(\mathbf{x})}{\sigma_{\widehat{G}}(\mathbf{x})}\right) - \phi\left(\frac{(a-\epsilon)-\widehat{G}(\mathbf{x})}{\sigma_{\widehat{G}}(\mathbf{x})}\right) - \phi\left(\frac{(a+\epsilon)-\widehat{G}(\mathbf{x})}{\sigma_{\widehat{G}}(\mathbf{x})}\right)\right] \\ &+ \left[\Phi\left(\frac{(a+\epsilon)-\widehat{G}(\mathbf{x})}{\sigma_{\widehat{G}}(\mathbf{x})}\right) - \Phi\left(\frac{(a-\epsilon)-\widehat{G}(\mathbf{x})}{\sigma_{\widehat{G}}(\mathbf{x})}\right)\right]\end{aligned} \tag{14}$$

where Φ is the standard normal cumulative distribution function and ϕ the standard normal density function.

In the case of reliability, the threshold a is 0. In EGRA, the expected feasibility function is built with $\epsilon = 2\sigma^2_{\widehat{G}}$. The same ϵ is selected for AK-MCS+EFF.

Table 1
Definition of the learning criterion (Stage 5) and the stopping condition (Stage 6) for the two learning functions proposed. They are given for $\mathbf{x} \in S$.

Learning function	EFF	U
Learning criterion	$\max(\mathrm{EFF}(\mathbf{x}))$	$\min(U(\mathbf{x}))$
Stopping condition	$\max(\mathrm{EFF}(\mathbf{x})) \leqslant 0.001$	$\min(U(\mathbf{x})) \geqslant 2$

Stage 5 of AK-MCS, the best next point $\mathbf{x}^*$ in S to evaluate has to be identified by the learning criterion. It is mentioned Section 3.2 that it depends on the choice of the learning function. In the case of EFF, the best next point to evaluate is the point among S having the maximum EFF value [5]. The learning criterion is then $\max(\mathrm{EFF}(\mathbf{x}))$ for $\mathbf{x} \in S$.

Stage 6, the learning function value of $\mathbf{x}^*$ is compared to a stopping one. In [5], it is defined as 0.001. Therefore, once the maximum of the expected feasibility function for the points in S is below 0.001, AK-MCS stops the learning and goes to Stage 8. Table 1 sums up the different parameters for EFF, *i.e.* the learning criterion and the stopping criterion.

EFF reflects a balance trend between the search in the vicinity of the limit state and a more global search in the design space [5]. It is important to note that this learning function is designed for EGRA that is to say for a method that approximates the whole limit state. Here, we classify points which are obtained from a Monte Carlo population. It is a different concept which does not guarantee EFF to perform well with AK-MCS. For this reason, we propose the more suitable learning function that we name U.

3.3.2. Learning function U

Coming from the statement that in Monte Carlo Simulations only the sign of the performance function is important, a learning function based on a different concept than EFF can be proposed. Indeed, in AK-MCS, accuracy on the sign is needed only among the points of the Monte Carlo population and the limit state can be roughly approximated in configurations with low densities of probabilities according to the distributions of the random variables. The points with a high potential risk of crossing the predicted separator $\widehat{G}(\mathbf{x}) = 0$ have to be added to the design of experiments and therefore evaluated on the performance function. In fact, the uncertainty on these points can cause them to change from positive to negative (or negative to positive) predicted values. This leads then to a modification of the probability of failure. The potentially "dangerous" points can present three characteristics: to be close to the limit state, to have an important uncertainty (high Kriging variance) or both at the same time. To identify them, the learning function $U(\mathbf{x})$ is proposed:

$$|\widehat{G}(\mathbf{x})| - U(\mathbf{x})\ \sigma_{\widehat{G}}(\mathbf{x}) = 0 \tag{15}$$

It indicates the distance in Kriging standard deviations between the prediction and the estimated limit state. It represents a reliability index on the risk of making a mistake on the sign of $G(\mathbf{x})$ considering $G(\mathbf{x})$ with the same sign than $\widehat{G}(\mathbf{x})$. This index $U(\mathbf{x})$ can be related to the lower confidence bounding (lcb) function proposed by Cox and John [23] for optimisation. In fact, they define lcb as:

$$lcb(\mathbf{x}) = \widehat{G}(\mathbf{x}) - b\ \sigma_{\widehat{G}}(\mathbf{x}) \tag{16}$$

First, they select b as equal to 2 or 2.5. This choice depends on the wish of a local improvement $b = 2$ or a more global one $b = 2.5$. Then, lcb is computed and minimised. Here, it works the opposite way as it is a reliability problem and not an optimisation one. lcb is not computed as only the sign of the prediction matters, therefore it is defined as $lcb = 0$. Then, $U(\mathbf{x})$ which is equivalent to b

in (16) is computed (Stage 5) on the whole population. The best point to evaluate becomes the one in S which minimises $U(\mathbf{x})$. The learning criterion (Stage 5) is then $\min(U(\mathbf{x}))$ for $\mathbf{x} \in S$.

Stage 6 requires the stopping condition. For U, it is defined as $\min(U(\mathbf{x})) \geqslant 2$ for $\mathbf{x} \in S$. As a reliability index, $U = 2$ corresponds to a probability of making a mistake on the sign of $\Phi(-2) = 0.023$. Cox and John [23] propose this value of 2 for their optimisation algorithm and moreover, it is seen on the following validation examples that such a stopping condition guarantees great accuracy. Table 1 sums up the learning parameters for U.

The learning function U is seen to give more weight to the points in the close neighbourhood of the predicted limit state rather than further ones with high Kriging variance like EFF can do. This learning function is therefore slower than EFF in estimating roughly the probability of failure but faster in converging towards an accurate probability of failure at the stopping condition. The choice of this learning function U is rather natural. It is easy to understand that the U value is directly linked to the Kriging variance and prediction. Aside from its use in optimisation in a different way, a close idea is proposed by Juang et al. [24] in the science of environment to improve the reliability of Kriging-based delineation of contaminated soils.

4. AK-MCS academic validation

AK-MCS efficiency is illustrated on several examples which cover a wide variety of limit states: first, examples of dimension 2 are tested to observe the method's behaviour. They were selected for their high non-linearity and rather complex limit state. Following this, a non-linear analytical function with a moderate dimension (<10) is tested. It corresponds to the dynamic response of a non-linear oscillator. To finish, the method is performed on a high dimension (40,100) analytical example. These examples are compared to crude Monte Carlo Simulation and results from literature.

4.1. Example 1: series system with four branches

The first example consists of a series system with four branches which has been proposed in [25–27]. The random variables x_i are standard normal distributed random variables. The performance function reads:

$$G(x_1, x_2) = \min \left\{ \begin{array}{l} 3 + 0.1(x_1 - x_2)^2 - \frac{(x_1+x_2)}{\sqrt{2}}; \\ 3 + 0.1(x_1 - x_2)^2 + \frac{(x_1+x_2)}{\sqrt{2}}; \\ (x_1 - x_2) + \frac{k}{\sqrt{2}}; (x_2 - x_1) + \frac{k}{\sqrt{2}} \end{array} \right\} \quad (17)$$

with k taking two values (6 then 7) to fit literature examples.

AK-MCS is compared with numerous metamodels proposed in [26,27]. The results are summarised Tables 2 and 3. The information is given as the number of calls to the performance function N_{call}, the number of points n_{MC} in the Monte Carlo population S, the probability of failure, the corresponding reliability index and the error percentage compared to the reference reliability index. The proposed method called AK-MCS is tested with the two different learning functions defined previously (U and EFF) on the same Monte Carlo population and initial design of experiments.

4.1.1. Reliability results

First, the results (Tables 2 and 3) show that the AK-MCS is more efficient than the other metamodels found in literature. Indeed, the number of calls to the performance function is lower than most of the metamodels proposed in [26,27] and the prediction of the probability of failure is very accurate. For $k = 6$ (Table 2), only crude Directional Sampling and Importance Sampling + Neural Network (IS+NN) require less calls to the performance function. However,

Table 2
Example 1, $k = 6$, reliability results – comparison of AK-MCS (population size: $n_{MC} = 10^6$) with a Monte Carlo Simulation on the same population and with metamodels from [26]. Legend: N_{call}, number of calls to the performance function; $\widehat{P_f}$, the probability of failure; $\text{C.O.V}_{\widehat{P_f}}$, its coefficient of variation for the Monte Carlo Simulation; β, the corresponding reliability index and ϵ_β, its percentage error in comparison with the reference reliability index (the Monte Carlo Simulation result for AK-MCS and the reference value mentioned in [26] for the other metamodels). If the predicted $\widehat{P_f}$ and β are similar to the reference value, $*$ is mentioned in the last column.

Method	N_{call}	$\widehat{P_f}$ ($\text{C.O.V}_{\widehat{P_f}}$)	β	ϵ_β (%)
Monte Carlo	10^6	4.416×10^{-3} (1.5%)	2.618	–
AK-MCS+*U*	126	4.416×10^{-3}	2.618	$*$
AK-MCS+EFF	124	4.412×10^{-3}	2.619	0.004
Directional Sampling (DS)	52	4.5×10^{-3}	2.61	0.04
DS + Response Surface	1745	5.0×10^{-3}	2.57	1.91
DS + Spline	145	2.4×10^{-3}	2.82	7.63
DS + Neural Network	165	4.1×10^{-3}	2.64	0.76
Importance Sampling (IS)	1469	4.9×10^{-3}	2.58	1.53
IS + Response Surface	1375	4.5×10^{-3}	2.62	$*$
IS + Spline	428	4.5×10^{-3}	2.62	$*$
IS + Neural Network	52	5.7×10^{-3}	2.53	3.44

Table 3
Example 1, $k = 7$, reliability results – comparison of AK-MCS (population size: $n_{MC} = 10^6$) with a Monte Carlo Simulation on the same population and with metamodels from [27]. Legend: N_{call}, number of calls to the performance function; $\widehat{P_f}$, the probability of failure; $\text{C.O.V}_{\widehat{P_f}}$, its coefficient of variation for the Monte Carlo Simulation; β, the corresponding reliability index and ϵ_β, its percentage error in comparison with the reference reliability index (the Monte Carlo Simulation result for AK-MCS and the reference value mentioned in [27] for the other metamodels). If the predicted $\widehat{P_f}$ and β are similar to the reference value, $*$ is mentioned in the last column.

Method	N_{call}	$\widehat{P_f}$ ($\text{C.O.V}_{\widehat{P_f}}$)	β	ϵ_β (%)
Monte Carlo	10^6	2.233×10^{-3} (2.1%)	2.843	–
AK-MCS+*U*	96	2.233×10^{-3}	2.843	$*$
AK-MCS+EFF	101	2.232×10^{-3}	2.843	$<10^{-4}$
Directional Sampling (DS)	9192	2.6×10^{-3}	2.79	2.11
DS + Response Surface	830	1.0×10^{-3}	3.03	6.32
DS + Spline	107	1.0×10^{-3}	3.00	5.26
DS + Neural Network	67	1.0×10^{-3}	3.05	7.01
DS + Ordinary Kriging	107	1.5×10^{-3}	2.95	3.51
Importance Sampling (IS)	4750	2.2×10^{-3}	2.84	0.35
IS + Response Surface	3877	2.0×10^{-3}	2.84	0.35
IS + Spline	724	2.0×10^{-3}	2.81	1.40
IS + Neural Network	125	2.9×10^{-3}	2.76	3.16
IS + Ordinary Kriging	146	2.0×10^{-3}	2.88	1.05

the nearly radial shape of the limit state is an adequate problem to solve with crude directional sampling [26]. About IS+NN, it must be seen that the reliability index is not very accurate. An error of 3% is obtained in comparison to the reference reliability index [26]. The same remark can be made for the second application with $k = 7$ (Table 3), where Directional Sampling + Neural Network (DS+NN) needs less calls but remains less accurate (7% error on the reliability index). A method with Ordinary Kriging (OK) is also proposed in [27] for $k = 7$. This method is found to require more calls to the performance function than AK-MCS and its accuracy remains very poor too.

AK-MCS shows great effectiveness and it must be noticed that AK-MCS+*U* gives a prediction of the probability of failure similar to the one obtained by Monte Carlo Simulation on the true performance function for the same population for both k. This can be linked to Fig. 2 which shows that an accurate separator is defined in regions, where Monte Carlo sampling points are situated. Locations with extremely weak densities of probability are badly approximated as they do not present any interest in the

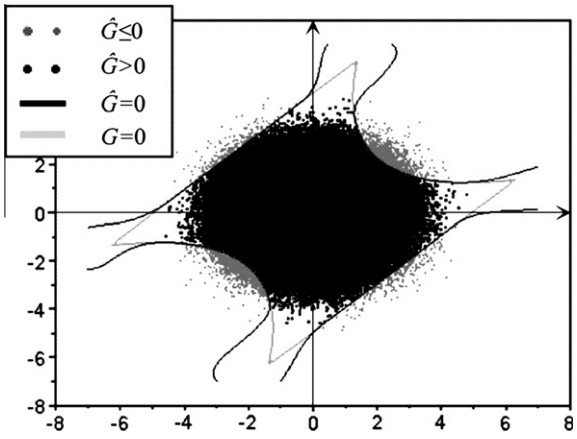


Fig. 2. Example 1, Monte Carlo population estimated by AK-MCS+*U* for $k = 7$.

computation of the probability of failure. Consequently, this method certifies a correct approximation of the probability of failure with a minimum number of calls to the performance function.

4.1.2. Comparison of the learning functions

According to the results, the choice of the learning function does not seem to have a real impact on this example. However, the use of the function *U* tends to give better results as the probability of failure is similar to the one estimated by Monte Carlo Simulation on the performance function. AK-MCS+EFF is found to give a slightly less accurate probability than AK-MCS+*U* for approximately the same number of calls to the performance function. To improve accuracy, the stopping condition has to be more restrictive but this would require more evaluations of *G*. By plotting the evolution of the probability of failure in terms of the number of calls to the performance function for $k = 7$ (Fig. 3), it can be seen that AK-MCS+EFF converges earlier towards the right probability. AK-MCS+*U* is slower to converge but the fastest to satisfy its stopping condition. Four levels (one for each branch) can be observed in its evolution Fig. 3. These levels are coming from the way the learning criterion works. Indeed, as the limit state has 4 different branches, the learning criterion of *U* focuses on one of them first. Once the metamodel is accurate enough in this region, the learning criterion goes to another branch and carries on.

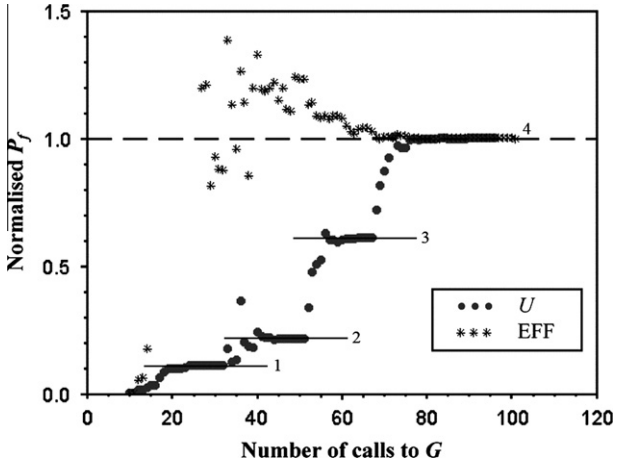


Fig. 3. Example 1, evolution of $\widehat{P_f}$ normalised by P_f obtained by Monte Carlo Simulation in terms of N_{call} for $k = 7$. Each level 1–4 corresponds to the learning step of *U* for each branch of the limit state.

For both learning functions, the probability of failure is well approximated with only 70 calls to the performance function for $k = 7$, the 30 following calls being only required to satisfy the stopping condition (Fig. 3). Depending on the aim of the analysis, it is possible to select a stopping condition like $U > 2$ or a more "visual" one by plotting the evolution of the probability of failure in terms of the number of calls to the performance function. However, this is riskier as for this example, one using AK-MCS+*U* could have been tempted to stop at the first branch that is to say about 30 calls.

To finish, the positions of the added points to the design of experiments can be compared for *U* and EFF. It is seen that the learning criterion of EFF is likely to add more points in regions which are not of interest (Fig. 4), whereas for *U*, it really focuses on the limit state (Fig. 5).

4.2. Example 2: modified Rastrigin function

The next example consists of a modified version of the Rastrigin function [28]. This modification is made to obtain negative (failure) and positive (safe) values. The idea, here, is to define a highly nonlinear performance function involving non-convex and non-connex domains of failure (*i.e.* "scattered gaps of failure"). The random variables x_i are standard normal distributed random variables. The performance function reads:

$$G(x_1, x_2) = 10 - \sum_{i=1}^{2} \left(x_i^2 - 5\cos(2\pi x_i)\right) \tag{18}$$

The method is compared Table 4 with Monte Carlo Simulation, Subset Simulations [29] performed with Phimeca-Soft [30] and a passive Kriging metamodel, that is based on a fixed Latin Hypercube design of experiments which size is equal to the number of calls required by AK-MCS+*U*.

This example shows that AK-MCS can be used on highly nonlinear limit states and on problems involving non-convex and non-connex domains of failure. Its performance is incomparable to Subset Simulation or even to passive Kriging combined with Latin Hypercube sampling. Indeed, the probability of failure estimated by AK-MCS with only 400 calls is found to be the same than the one obtained by Monte Carlo Simulation for the same population (Table 4). Furthermore, in this case, AK-MCS+*U* and AK-MCS+EFF perform the same. However, it must be noticed Fig. 6 that AK-MCS+EFF gives an earlier convergence than AK-MCS+*U*. Fig. 7 shows the design of experiments required by AK-MCS+*U* to satisfy the stopping condition and the similar probability of failure than the one estimated by a classic Monte Carlo Simulation on the same population. It is seen that the points are

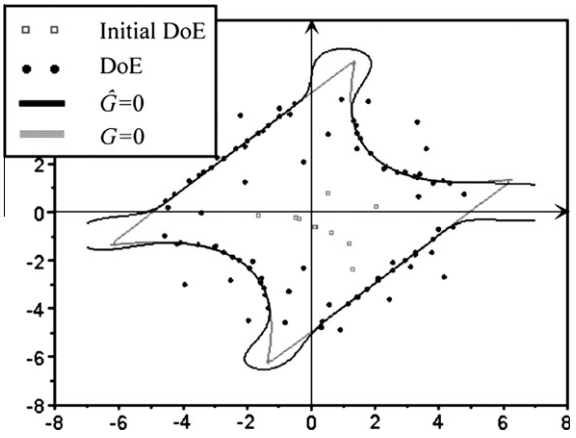


Fig. 4. Example 1, approximation by AK-MCS+EFF for $k = 7$.

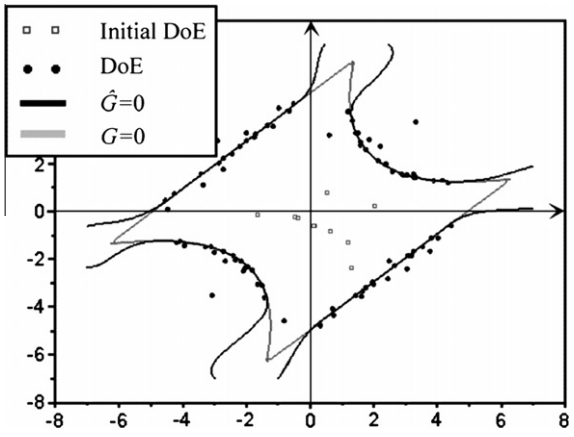


Fig. 5. Example 1, approximation by AK-MCS+U for $k = 7$.

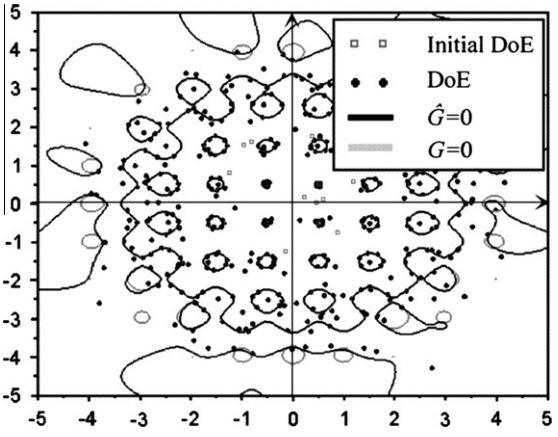


Fig. 7. Example 2, approximation by AK-MCS+U.

Table 4
Example 2, reliability results – comparison of AK-MCS (population size: $n_{MC} = 6 \times 10^4$) with a Monte Carlo Simulation on the same population, with passive Kriging and Latin Hypercube Sampling and Subset Simulation using Phimeca-Soft [30]. Legend: N_{call}, number of calls to the performance function (expensive); $\widehat{P_f}$, the probability of failure; $\mathrm{C.O.V}_{\widehat{P_f}}$, its coefficient of variation for the Monte Carlo Simulation; β, the corresponding reliability index and ϵ_β, its percentage error in comparison with the reference reliability index (Monte Carlo Simulation result). If the predicted $\widehat{P_f}$ and β are similar to the reference value, $*$ is mentioned in the last column.

Method	N_{call}	$\widehat{P_f}$ ($\mathrm{C.O.V}_{\widehat{P_f}}$)	β	ϵ_β (%)
Monte Carlo	6×10^4	7.34×10^{-2} (1.5%)	1.45	–
AK-MCS+U	416	7.34×10^{-2}	1.45	$*$
AK-MCS+EFF	417	7.34×10^{-2}	1.45	$*$
Passive Kriging+LHS	416	6.73×10^{-2}	1.50	3.44
Subset	5000	7.65×10^{-2}	1.43	1.45

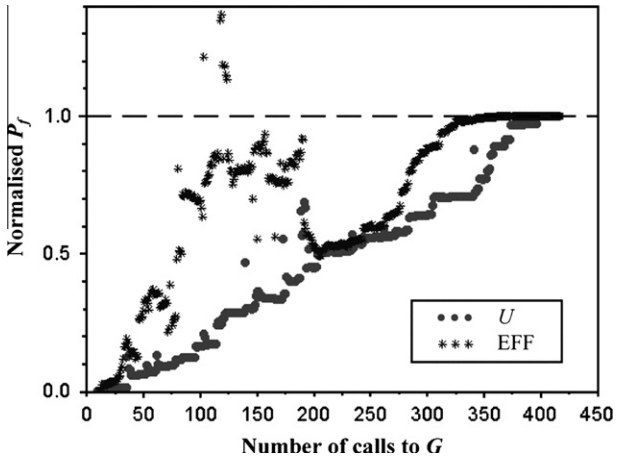


Fig. 6. Example 2, evolution of $\widehat{P_f}$ normalised by P_f obtained by Monte Carlo Simulation in terms of N_{call}.

concentrated in the vicinity of the limit state. Furthermore, Fig. 8 shows that the limit state is well approximated, where the Monte Carlo population is located. Indeed, the approximation coincides exactly with the limit state in the region covered by the Monte Carlo population. Then, the interpolation worsens, where no Monte Carlo points are observed. Once again, this is not important as it corresponds to locations which present extremely weak densities of probability and therefore have a negligible effect on the probability of failure.

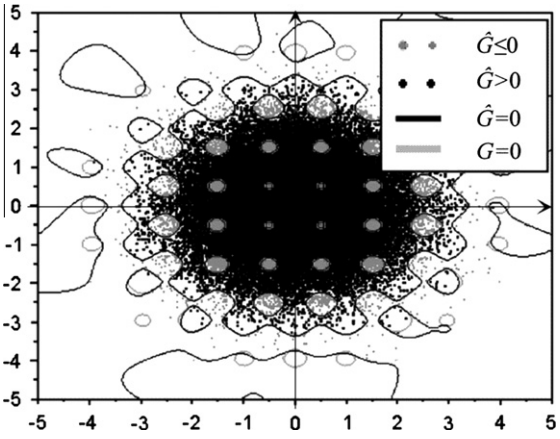


Fig. 8. Example 2, Monte Carlo population estimated by AK-MCS+U.

4.3. Example 3: dynamic response of a non-linear oscillator

The following example is a problem with a moderate number of random variables. It consists of a non-linear undamped single degree of freedom system (Fig. 9). This example is also studied in [26,31,32]. The performance function is given as:

$$G(c_1, c_2, m, r, t_1, F_1) = 3r - \left| \frac{2F_1}{m\,\omega_0^2} \sin\left(\frac{\omega_0^2\, t_1}{2}\right) \right| \tag{19}$$

with $\omega_0 = \sqrt{\frac{c_1+c_2}{m}}$. The six random variables are listed Table 5. AK-MCS is compared with a Monte Carlo Simulation and

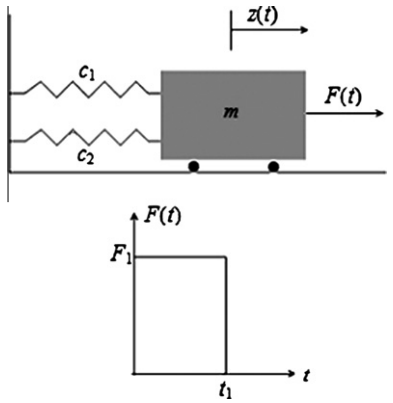


Fig. 9. Example 3, non-linear oscillator [26].

Table 5
Example 3, random variables [26].

Variable	P.D.F.	Mean	Standard deviation
m	Normal	1	0.05
c_1	Normal	1	0.1
c_2	Normal	0.1	0.01
r	Normal	0.5	0.05
F_1	Normal	1	0.2
t_1	Normal	1	0.2

metamodels found in literature. It is seen Table 6 that AK-MCS requires less calls to the performance function than the metamodels proposed in [26]. Furthermore, the probability of failure of AK-MCS+*U* is found to be similar to the one estimated by Monte Carlo Simulation on the true performance function with the same population, whereas AK-MCS+EFF, which requires less calls to reach its stopping condition, is less accurate.

The performance function is well approximated with about 30 calls for the two learning functions (Fig. 10). Among other strategies proposed in [26], only DS+NN is found to be as accurate as AK-MCS. However, it requires more calls to obtain the right probability of failure. This example shows that AK-MCS can be applied on moderate dimension problems with great effectiveness.

4.4. Example 4: high dimensional example

The last example is proposed in [33]. It consists of an analytical performance function, where the number of variables can be changed without modifying significantly the level of failure probability. It reads:

$$G(x_1,\ldots,x_n) = (n + 3\sigma\sqrt{n}) - \sum_{i=1}^{n} x_i \tag{20}$$

The random variables are taken as lognormal variables with unit means and standard deviation of $\sigma = 0.2$. Two studies are performed on this example: 40 and 100 random variables. The results of the reliability analyses are summarised Table 7 and are compared with Monte Carlo Simulations of 3×10^5 points. Less than 200 calls to the performance function are required for both AK-MCS methods to obtain the similar probability of failure than the Monte Carlo one on the same population. It is shown that on this example, increasing the number of random variables on the same level of failure has little influence on the number of calls required.

Table 6
Example 3, reliability results – comparison of AK-MCS (population size: $n_{MC} = 7 \times 10^4$) with a Monte Carlo Simulation on the same population and with metamodels from [26]. Legend: N_{call}, number of calls to the performance function; $\widehat{P_f}$, the probability of failure; $\mathrm{C.O.V}_{\widehat{P_f}}$, its coefficient of variation for the Monte Carlo Simulation; β, the corresponding reliability index and ϵ_β, its percentage error in comparison with the reference reliability index (the Monte Carlo Simulation result for AK-MCS and the reference value mentioned in [26] for the other metamodels). If the predicted $\widehat{P_f}$ and β are similar to the reference value, $*$ is mentioned in the last column.

Method	N_{call}	$\widehat{P_f}$ ($\mathrm{C.O.V}_{\widehat{P_f}}$)	β	ϵ_β (%)
Monte Carlo	7×10^4	2.834×10^{-2} (2.2%)	1.906	–
AK-MCS+*U*	58	2.834×10^{-2}	1.906	$*$
AK-MCS+EFF	45	2.851×10^{-2}	1.903	0.16
Directional Sampling (DS)	1281	3.5×10^{-2}	1.81	5.24
DS + Response Surface	62	3.4×10^{-2}	1.82	4.71
DS + Spline	76	3.4×10^{-2}	1.83	4.19
DS + Neural Network	86	2.8×10^{-2}	1.91	$*$
Importance Sampling (IS)	6144	2.7×10^{-2}	1.93	1.04
IS + Response Surface	109	2.5×10^{-2}	1.96	2.62
IS + Spline	67	2.7×10^{-2}	1.93	1.04
IS + Neural Network	68	3.1×10^{-2}	1.87	2.01

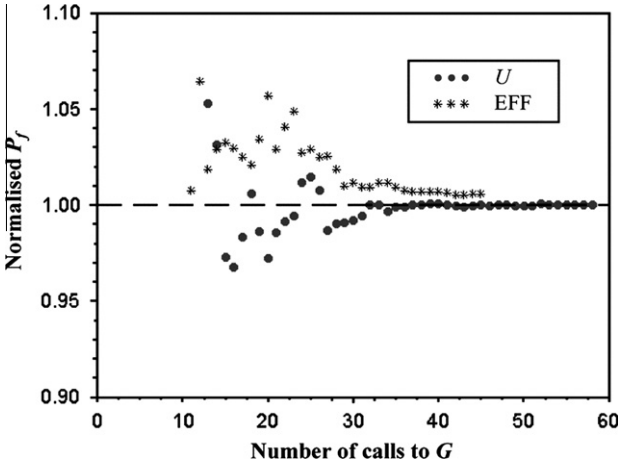


Fig. 10. Example 3, evolution of $\widehat{P_f}$ normalised by P_f obtained by Monte Carlo Simulation in terms of N_{call}.

Table 7
Example 4, reliability results - Comparison of AK-MCS (population size: $n_{MC} = 3 \times 10^5$) with a Monte Carlo Simulation on the same population. Legend: N_{call}, number of calls to the performance function; $\widehat{P_f}$, the probability of failure; $\mathrm{C.O.V}_{\widehat{P_f}}$, its coefficient of variation for the Monte Carlo Simulation; β, the corresponding reliability index.

Method	n	N_{call}	$\widehat{P_f}$ ($\mathrm{C.O.V}_{\widehat{P_f}}$)	β
Monte Carlo	40	3×10^5	1.813×10^{-3} (4.3%)	2.91
	100	3×10^5	1.647×10^{-3} (4.5%)	2.94
AK-MCS+*U*	40	112	1.813×10^{-3}	2.91
	100	153	1.647×10^{-3}	2.94
AK-MCS+EFF	40	124	1.813×10^{-3}	2.91
	100	167	1.647×10^{-3}	2.94

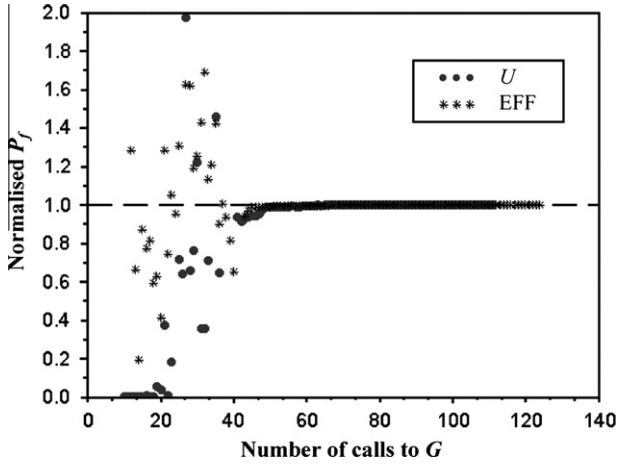


Fig. 11. Example 4, evolution of $\widehat{P_f}$ normalised by P_f obtained by Monte Carlo Simulation in terms of N_{call} for $n = 40$.

For 40 random variables, the probability of failure starts to converge around 50 calls to the performance function. AK-MCS+EFF and AK-MCS+*U* show the same behaviour in this example and none seems to give a faster convergence (Fig. 11). The initial design of experiments of small size is found to perform well. Indeed, all efforts are concentrated on adding interesting points rather than supplying the metamodel with a large initial design of experiments to cover space like in [5]. Furthermore, the use of a Monte Carlo population enables to reduce the number of points to predict. In fact, to approximate sufficiently the performance function with

the other methods presented before, a large number of predictions is needed to cover space. Here, once again, the use of a Monte Carlo population is very relevant as it focuses only on configurations with sufficiently high densities of probability.

5. Conclusion

This paper proposes an Active learning method combining Kriging and Monte Carlo Simulation (AK-MCS) to perform structural reliability analysis. The proposed strategy is found to be economic in the number of calls to the expensive performance function and its results are very accurate for the probability of failure. This approach combines the advantages of two methods: Kriging metamodel and Monte Carlo simulation. The idea is to perform a Monte Carlo simulation without evaluating the whole population. Indeed, the population is predicted using a Kriging metamodel which is defined thanks to only a few points of the population that are evaluated on the performance function. This strategy permits to focus on configurations with sufficiently high densities of probability. Therefore, the performance function's sign is not approximated accurately in the whole design space but only, where Monte Carlo points are located. Moreover, the active learning is preferred to update the Kriging metamodel by adding (and evaluating on G) a new point step by step to the design of experiments. This is done thanks to the Kriging variance and prediction which permit to define a learning function to target the best next point to evaluate, that is to say the point which is likely to cross the predicted limit state and to disrupt the probability of failure for the learning function U. This results in reducing drastically the number of calls to the performance function (and therefore the computation time) without penalising the accuracy on the probability of failure. Two learning functions are compared (U,EFF). AK-MCS+U is found to perform slightly better in most of the cases. To validate the approach, the method is conducted on several examples which cover a wide variety of limit states: high non-linearity, non-convex and non-connex domains of failure, moderate dimensional problem (<10) and high dimensional one. AK-MCS performs well in all these cases as the number of calls to the performance function is found to be small and the accuracy on the probability of failure very high. Even if some approaches such as Directional Sampling can be more efficient for suitable limit states such as the first example with $k = 6$, AK-MCS is adapted to singularities. Its ability to adapt and its parsimony open a new way to applications implying Finite Element mechanical model. Furthermore, it is seen on the high dimensional example that the number of calls to the expensive performance function is not really sensitive to the number of random variables. The method can be performed either in the physical space regardless of the distributions or in the standard space by converting the random variables to standard normal ones. Indeed, as underlined in [5], Kriging is not greatly affected by the choice of the space as no assumptions are made on the shape of the limit state. Moreover, the use of a Monte Carlo population enables to perform AK-MCS on problems for which the topology is not known. To finish, on the example of AK-MCS, some other methods can be proposed such as AK–IS or AK-SUBSET which define an Active learning method combining Kriging with respectively Importance Sampling or Subset Simulation. These methods open the way to efficient calculations of very weak probabilities of failure.

Acknowledgements

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